

# Data Structures Homework - 1

## 1.Sum of Two Matrices

```
#include <stdio.h>

int main() {
    int r, c, i, j;
    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);

    int a[r][c], b[r][c], sum[r][c];

    printf("Enter first matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&b[i][j]);

    printf("Sum Matrix:\n");
    for(i=0;i<r;i++) {
        for(j=0;j<c;j++) {
            sum[i][j]=a[i][j]+b[i][j];
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

### Output :

```
Enter rows and columns: 3
3
Enter first matrix:
12
23
```

34

45

56

67

8

8

7

Enter second matrix:

23

32

43

4

65

76

87

98

6

Sum Matrix:

35 55 77

49 121 143

95 106 13

=== Code Execution Successful ===

## 2.Reverse an Array

```
#include <stdio.h>
```

```
int main() {
```

```
    int n,i;
```

```
    printf("Enter size: ");
```

```
    scanf("%d",&n);
```

```
    int a[n];
```

```
    printf("Enter elements:\n");
```

```
    for(i=0;i<n;i++)
```

```
        scanf("%d",&a[i]);
```

```
    printf("Reverse Order:\n");
```

```
    for(i=n-1;i>=0;i--)
```

```
        printf("%d ",a[i]);
```

```
    return 0;
}
```

Output :

Enter size: 5

Enter elements:

23

45

56

7

8

Reverse Order:

8 7 56 45 23

=== Code Execution Successful ===

### 3.Count Duplicate Elements

```
#include <stdio.h>
```

```
int main() {
    int n,i,j,count=0;
    printf("\n Enter the N value :");
    scanf("%d",&n);

    int a[n],check[n];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++) {
        scanf("%d",&a[i]);
        check[i]=0;
    }

    for(i=0;i<n;i++) {
        if(check[i]) continue;
        for(j=i+1;j<n;j++) {
            if(a[i]==a[j]) {
                check[j]=1;
                count++;
            }
        }
    }
}
```

```

    }

    printf("Duplicate Elements = %d",count);
    return 0;
}

```

Output :

```

Enter the N value :5
Enter the array Elements :
23
23
45
45
45
Duplicate Elements = 3

```

=== Code Execution Successful ===

#### 4.Print Unique Elements

```

#include <stdio.h>

int main() {
    int n,i,j,flag;
    printf("\n Enter the N value :");
    scanf("%d",&n);
    int a[n];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Unique Elements:\n");

    for(i=0;i<n;i++) {
        flag=1;
        for(j=0;j<n;j++) {
            if(i!=j && a[i]==a[j]) {
                flag=0;
                break;
            }
        }
    }
}

```

```

        if(flag)
            printf("%d ",a[i]);
    }
    return 0;
}

```

### Output :

```

Enter the N value :5
Enter the array Elements :
34
45
45
56
5
Unique Elements:
34 56 5

```

=== Code Execution Successful ===

### 5.Insert into Sorted Array

```

#include <stdio.h>

int main() {
    int n,i,pos,value;
    printf("\n Enter the N value :");
    scanf("%d",&n);

    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter value: ");
    scanf("%d",&value);

    for(i=0;i<n;i++) {
        if(a[i]>value)
            break;
    }
}

```

```

pos=i;

for(i=n;i>pos;i--)
    a[i]=a[i-1];

a[pos]=value;
n++;

for(i=0;i<n;i++)
    printf("%d ",a[i]);

return 0;
}

```

### Output :

```

Enter the N value :5
Enter the array Elements :
34
5
6
7
78
Enter value: 45
34 5 6 7 45 78

```

=== Code Execution Successful ===

### 6.Insert into Unsorted Array

```

#include <stdio.h>

int main() {
    int n,i,pos,value;
    printf("\n Enter the N value :");
    scanf("%d",&n);

    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
}

```

```

printf("Enter position: ");
scanf("%d",&pos);

printf("Enter value: ");
scanf("%d",&value);

for(i=n;i>=pos;i--)
    a[i]=a[i-1];

a[pos-1]=value;
n++;

for(i=0;i<n;i++)
    printf("%d ",a[i]);

return 0;
}

```

### Output :

```

Enter the N value :5
Enter the array Elements :
34
45
56
67
78
Enter position: 4
Enter value: 65
34 45 56 65 67 78

```

=== Code Execution Successful ===

### 7.Matrix Multiplication

```
#include <stdio.h>
```

```

int main()
{
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of first matrix: ");

```

```

scanf("%d%d", &r1, &c1);

printf("Enter rows and columns of second matrix: ");
scanf("%d%d", &r2, &c2);

if (c1 != r2)
{
    printf("Matrix multiplication is not possible.");
    return 0;
}

int a[r1][c1], b[r2][c2], c[r1][c2];

printf("Enter first matrix elements:\n");
for (i = 0; i < r1; i++)
{
    for (j = 0; j < c1; j++)
    {
        scanf("%d", &a[i][j]);
    }
}

printf("Enter second matrix elements:\n");
for (i = 0; i < r2; i++)
{
    for (j = 0; j < c2; j++)
    {
        scanf("%d", &b[i][j]);
    }
}

for (i = 0; i < r1; i++)
{
    for (j = 0; j < c2; j++)
    {
        c[i][j] = 0;
        for (k = 0; k < c1; k++)
        {
            c[i][j] += a[i][k] * b[k][j];
        }
    }
}

```



```

printf("Product Matrix:\n");

for (i = 0; i < r1; i++)
{
    for (j = 0; j < c2; j++)
    {
        printf("%d ", c[i][j]);
    }
    printf("\n");
}

return 0;
}

```

### **Output :**

Enter rows and columns of first matrix: 3

3

Enter rows and columns of second matrix: 3

3

Enter first matrix elements:

1

2

3

4

5

6

7

8

9

Enter second matrix elements:

9

8

7

6

5

4

3

2

1

Product Matrix:

```
30 24 18
84 69 54
138 114 90
```

=== Code Execution Successful ===

## 8.Matrix Transpose

```
#include <stdio.h>

int main()
{
    int r, c, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d%d", &r, &c);

    int a[r][c];

    printf("Enter matrix elements:\n");

    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Transpose Matrix:\n");

    for (i = 0; i < c; i++)
    {
        for (j = 0; j < r; j++)
        {
            printf("%d ", a[j][i]);
        }
        printf("\n");
    }

    return 0;
```

```
}
```

### Output :

Enter number of rows and columns: 3

3

Enter matrix elements:

11

22

33

44

55

6

77

88

99

Transpose Matrix:

11 44 77

22 55 88

33 6 99

=== Code Execution Successful ===

### 9.Right Diagonal Sum

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, i, j, sum = 0;
```

```
    printf("Enter the order of matrix: ");
```

```
    scanf("%d", &n);
```

```
    int a[n][n];
```

```
    printf("Enter matrix elements:\n");
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        for (j = 0; j < n; j++)
```

```

        {
            scanf("%d", &a[i][j]);
        }
    }

    for (i = 0; i < n; i++)
    {
        sum += a[i][n - 1 - i];
    }

    printf("Right diagonal sum = %d", sum);

    return 0;
}

```

Output :

Enter the order of matrix: 3

Enter matrix elements:

9

8

7

6

5

4

3

2

1

Right diagonal sum = 15

=== Code Execution Successful ===

## 10. Check Identity Matrix

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, i, j, flag = 1;
```

```
    printf("Enter the order of matrix: ");
```

```
    scanf("%d", &n);
```

```

int a[n][n];

printf("Enter matrix elements:\n");

for (i = 0; i < n; i++)
{
    for (j = 0; j < n; j++)
    {
        scanf("%d", &a[i][j]);
    }
}

for (i = 0; i < n; i++)
{
    for (j = 0; j < n; j++)
    {
        if (i == j && a[i][j] != 1)
        {
            flag = 0;
        }

        if (i != j && a[i][j] != 0)
        {
            flag = 0;
        }
    }
}

if (flag == 1)
    printf("It is an Identity Matrix.");
else
    printf("It is not an Identity Matrix.");

return 0;
}

```

### **Output :**

```

Enter the order of matrix: 3
Enter matrix elements:
1
2
3

```

2  
1  
4  
3  
4  
1

It is not an Identity Matrix.

=== Code Execution Successful ===

## 11.Count Vowels, Consonants, Digits and White Spaces

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char str[100];
```

```
    int i, vowels = 0, consonants = 0, digits = 0, spaces = 0;
```

```
    printf("Enter a string: ");
```

```
    scanf("%s", str);
```

```
    for (i = 0; str[i] != '\0'; i++)
```

```
    {
```

```
        if ((str[i] >= 'A' && str[i] <= 'Z') || (str[i] >= 'a' && str[i] <= 'z'))
```

```
        {
```

```
            char ch = str[i];
```

```
            if (ch >= 'A' && ch <= 'Z')
```

```
                ch = ch + 32;
```

```
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
```

```
                vowels++;
```

```
            else
```

```
                consonants++;
```

```
        }
```

```
    else if (str[i] >= '0' && str[i] <= '9')
```

```
    {
```

```
        digits++;
```

```
    }
```

```
    else if (str[i] == ' ')
```

```
    {
```

```
        spaces++;
```

```

    }
}

printf("Vowels = %d\n", vowels);
printf("Consonants = %d\n", consonants);
printf("Digits = %d\n", digits);
printf("White Spaces = %d\n", spaces);

return 0;
}

```

### Output :

```

Enter a string: kanishkar
Vowels = 3
Consonants = 6
Digits = 0
White Spaces = 0

```

=== Code Execution Successful ===

### 12.Frequency of a Character

```

#include <stdio.h>

int main()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    scanf("%s", str);

    printf("Enter the character: ");
    scanf(" %c", &ch);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
        {
            count++;
        }
    }
}

```

```

    }

    printf("Frequency of %c = %d", ch, count);

    return 0;
}

```

### **Output :**

```

Enter a string: onepiece
Enter the character: e
Frequency of e = 3

```

=== Code Execution Successful ===

### 13.Count Words in a String

```

#include <stdio.h>

int main()
{
    char str[100];
    int i, words = 1;

    printf("Enter a string: ");
    scanf("%[^\\n]", str);

    for(i = 0; str[i] != '\\0'; i++)
    {
        if(str[i] == ' ')
        {
            words++;
        }
    }

    printf("Number of words = %d", words);

    return 0;
}

```

### **Output :**

```

Enter a string: hi im kanishkar

```



Number of words = 3

=== Code Execution Successful ===

#### 14.Sort String Array

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    int n, i, j;
```

```
    char str[20][50], temp[50];
```

```
    printf("Enter the number of strings: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter the strings:\n");
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        scanf(" %[^\\n]", str[i]);
```

```
    }
```

```
    for(i = 0; i < n - 1; i++)
```

```
    {
```

```
        for(j = i + 1; j < n; j++)
```

```
        {
```

```
            if(strcmp(str[i], str[j]) > 0)
```

```
            {
```

```
                strcpy(temp, str[i]);
```

```
                strcpy(str[i], str[j]);
```

```
                strcpy(str[j], temp);
```

```
            }
```

```
        }
```

```
    }
```

```
    printf("Strings in Sorted Order:\n");
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        printf("%s\\n", str[i]);
```

```
    }
```

```
    return 0;
}
```

### **Output :**

```
Enter the number of strings: 5
Enter the strings:
mani
akil
bala
sri
thiru
Strings in Sorted Order:
akil
bala
mani
sri
thiru
```

=== Code Execution Successful ===

### 15. Toggle Case of a Sentence

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;

    printf("Enter a sentence: ");
    scanf("%s", str);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
        else if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
    }
}
```

```
    }  
}  
  
printf("Sentence after toggling:\n%s", str);  
  
return 0;  
}
```

**Output :**

```
Enter a sentence: KaNiShKaR  
Sentence after toggling:  
kAnIsHkAr
```

=== Code Execution Successful ===